

Dajun Lin

Ph.D. Candidate, Electrical Engineering – University of Utah
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PROFESSIONAL SUMMARY

Ph.D. Candidate specializing in Single-shot Volumetric Holographic Lithography/Printing and End-to-end Wide FOV Achromatic Hybrid Optics. Expert in Nanofabrication, Optics + AI, Computational Photography, Inverse design problems, Differentiable ray-tracing, Holography, AR and VR optics.

EDUCATION

- **University of Utah** Salt Lake City, USA
Ph.D. in Electrical Engineering (Dr. Rajesh Menon group) Aug 2022 - Oct 2026 (expected)
- **University of Shanghai for Science and Technology** Shanghai, China
M.Eng. in Optics Engineering (Dr. Min Gu group) Sep 2018 - May 2021
- **Changzhou Institute of Technology** Changzhou, China
B.Eng. in Optics Engineering Sep 2014 - May 2018

WORKING EXPERIENCE

- **Futurewei Technologies**
Photonic and thermal simulation intern, IC Lab May 2026 - Aug 2026
 - **Differentiable ray-tracing refractive lens and Metalens modeling**
 - * Develop efficient and accurate optical modeling using Monte Carlo differentiable ray tracing and Huygens wave propagation for optical surfaces and diffractive elements.
 - * End-to-end curriculum-learning-based optimization of an 80° FOV achromatic refractive–diffractive hybrid lens incorporating a Meta-lens, with reduced lateral chromatic aberration, distortion, and field curvature.
 - **Conducted technical reviews of co-packaged optics (CPO) and near-packaged optics (NPO) architectures, key technologies, and industry trends**

PROJECT EXPERIENCE

- **Computational Imaging and Wide-FOV Achromatic Optics**
Research Assistant, Dr. Rajesh Menon group Aug 2022 - Present
 - **Wide-angle achromatic hybrid optics and neural image reconstruction**
 - * Build end-to-end differentiable ray-tracing and wave-propagation model for refractive-diffractive lenses.
 - * Introduced dispersion- and field-curvature-engineered hybrid refractive–diffractive designs incorporating quadratic-phase diffractive and metalens elements.
 - * Develop large-scale 135-mm f/4 and 20-mm f/4 full-frame camera lenses, and 100°-FOV hybrid lens for VR.
 - * Jointly optimized refractive–diffractive optics and neural reconstruction using an image-domain loss.
 - **Miniature multi-level diffractive doublet achieving 100° FOV imaging**
 - * Build an angular-spectrum (ASM) wave-based optimization model for wide-angle multi-plate flat lenses.
 - * Nanofabricate multi-level diffractive lenses using grayscale lithography with chromium-etched apertures and develop doublet alignment.
 - * Analyze experimental lens performance, including PSF, MTF, distortion, focusing efficiency, and vignetting.
- **Manufacturing USA: Cyber-enabled high-throughput multi-mat. 3D nanostruct.**
Research Assistant, Dr. Rajesh Menon group Jan 2023 – Present
 - **Single-exposure holographic 3D printing via inverse-designed phase masks**
 - * Develop ultra-fast, single-shot holographic printing of mm-scale 3D hollow structures without stitching.
 - * Develop inverse-designed computer-generated holography (CGH) phase masks, grayscale-lithography-based nanofabrication, engineered BAPO-based resin, and an exposure setup for volumetric polymerization.
 - **Single-shot volumetric photolithography producing high aspect-ratio microstructures**
 - * Overcomes the quasi-2D depth limitation of photolithography and the low throughput of serial point-writing.
 - * Develop a custom-built lithography, alignment, and monitoring system for high-aspect-ratio structures with shutter-controlled, stage-stepped, tilted, and overlapping exposures.
 - * Apply the structures to MEMS, capillary-driven flow, and in-situ SEM nanoindentation.
 - **Alignment-free**

- **Computational optics and photonic systems for communication, AI, and 3D display**
M.Eng. Thesis & Research Assistant, Dr. Min Gu group *Jan 2018 - Aug 2021*
 - **CNN-based LG-SK optics communication system**
 - * Implemented an spatial light modulator (SLM)-based optical setup for Laguerre–Gaussian beam and turbulence generation, with an automated data collection system.
 - * Build a CNN-based decoder for a 10-m, 768-ary LG-SK FSO system, achieving 95% accuracy, < 2.7 ms latency, and robust color image transmission.
 - **All-optical machine learning chips using diffractive neural networks**
 - * Build a machine-learning-based fully differentiable diffractive doublet for handwritten digit recognition.
 - * Fabricate a quartz-based bilayer chip with structures patterned on both sides of the wafer.
 - **3D holographic display of perovskite nanocrystals in glass**
 - * Fabricated tunable-bandgap nanocrystal structures through laser-induced nanophase separation.
 - * Develop 3D display algorithms and experimentally validate them using multiple aligned SLMs.
 - **Phase reconstruction & Wavefront imaging sensor**
 - * Developed a machine-learning-based single-shot phase retrieval framework for real-time imaging.
 - * Enabled high-res. phase imaging of biological cells & wavefronts under spectral broadband illumination.

SKILLS SUMMARY

- **Programming & Machine Learning & AI tools for coding and research**
 Python (TensorFlow, PyTorch, machine learning models), C++, MATLAB, LabVIEW.
- **Optics Photonics Design & Simulation**
 Zemax, Lumerical, COMSOL, RCWA, MEEP, CST, HFSS.
- **Micro/Nanofabrication**
 Photolithography, E-beam lithography (Raith Voyager), Two-photon polymerization (Nanoscribe), laser direct writing (DWL 66+), wet etching (Cr, HF/BOE), dry etching (ICP-RIE, Oxford 100), thin-film deposition (PVD, PECVD).
- **Metrology & Testing**
 8+ years of hands-on experience building optical setups, SEM, TEM, Keyence, Olympus LEXT confocal microscope.

SELECTED PUBLICATIONS

- [10] **Dajun Lin**, Brian Baker, et al. “Single-exposure holographic lithography of ultra-high aspect-ratio microstructures” *Nat. Commun* 17 (2026): 7175.
- [9] **Dajun Lin**, Xiaofeng Chen[#], et al. “Single-exposure holographic 3D printing via inverse-designed phase masks” *Sci. Adv.* 12.33 (2026): eaec3536.
- [8] **Dajun Lin**, Ping Lu, Apratim Majumder, Rajesh Menon. “Diffraction-free Orbital Angular Momentum Holography” *Opt. Express* (2025) 33 (3), 5791-5799.
- [7] Yibo Dong[#], **Dajun Lin**[#], et al. “Compact eternal diffractive neural network chip for extreme environments.” *Commun. Engineering* 3.1 (2024): 64.
- [6] **Dajun Lin**, Tina Hayward, Wei Jia, et al. “Inverse-designed multi-level diffractive doublet for wide field-of-view imaging” *ACS Photonics* 10.8 (2023): 2661-2669.
- [5] Jia Wei, **Dajun Lin**, and Berardi Sensale-Rodriguez. ”Machine learning enables multi-degree-of-freedom reconfigurable Terahertz holograms with cascaded diffractive optical elements.” *Adv. Opt. Materials* (2023): 2202538.
- [4] Dong Yibo, Haitao Luan, **Dajun Lin**, et al. ”Laser-Induced Graphene Hologram Reconfiguration for Countersurveillance Multisecret Sharing.” *Laser Photonics Rev.* (2023): 2200805.
- [3] Ping Lu, **Dajun Lin**, Ning Chen, et al. “CNN-assisted SERS enables ultra-sensitive and simultaneous detection of Scr and BUN for rapid kidney function assessment.” *Anal. Methods* (2023): 322-332.
- [2] Ke Sun, Dezhi Tan, Xinyuan Fang, Xintao Xia, **Dajun Lin**, et al. “Three-dimensional direct lithography of stable perovskite nanocrystals in glass.” *Science* 375.6578 (2022): 307-310.
- [1] Haitao Luan, **Dajun Lin**, Keyao Li, et al. “768-ary Laguerre–Gaussian-mode shift keying free-space optical communication based on convolutional neural networks.” *Opt. Express* 29.13 (2021): 19807-19818.

CONFERENCE & PRESENTATIONS

- [6, **Presenter**] Dajun Lin “Single-exposure volumetric photolithography producing ultra-high aspect-ratio microstructures” EIPBN Denver 2026
- [5, **Presenter**] Dajun Lin “Single-exposure holographic volumetric additive manufacturing” EIPBN Savannah 2025
- [4, **Presenter**] Dajun Lin “Single-exposure millimeter-scale volumetric holographic additive manufacturing” EIPBN La Jolla 2024
- [3, **Presenter**] Dajun Lin. “Diffraction-free orbital angular momentum holography enabled by an extended depth-of-focus flat lens” CLEO Charlotte 2024
- [2, **Presenter**] Dajun Lin. “Inch-scale Achromatic Planar Flat Lens Doublet Reshaping Near-eye Display” CLEO Charlotte 2024.
- [1, **Presenter**] Dajun Lin “Machine-learning-assisted inverse design for wide-FOV dual-layer multi-level diffractive lens integration” CLEO San Jose 2023

HONORS AND AWARDS

- 2025-2026 Graduate Research Fellowship \$28500, University of Utah. Awarded competitively to 10 fellows selected from 200 PhD applicants across the university for outstanding research potential.
- 2018-2021 Graduate Fellowship Award, University of Shanghai for Science and Technology

VOLUNTEER EXPERIENCE

- Session Student Helper, EIPBN Denver 2026, USA
- Session Student Helper, EIPBN Savannah 2025, USA
- Session Student Helper, EIPBN La Jolla 2024, USA